

MARSHALL BRUNER

☎ +1 719-553-9497 ✉ brunerm99@gmail.com  [marshall-bruner](https://www.linkedin.com/in/marshall-bruner)  [brunerm99](https://github.com/brunerm99)

EXPERIENCE

BAE Systems — DoD Clearance

Westminster, CO

RF Engineer

- System design, schematic, layout of 2D Active Phased Array Weather Radar
- Large scale phased array systems analysis ([Python](#), Matlab)
- Acted as a consultant for weather radar proposals

Ball Aerospace — DoD Clearance

Westminster, CO

RF Engineer

- Designed and debugged multiple complex, mission-critical RF boards under strict time constraints
- Simulated, designed, and tested an RF Front-End PCB at extreme temperatures
- Schematic capture and layout of RF T/R systems (RF, digital, power), antenna range test equipment, component testing coupons, etc.
- Built [data visualization and analysis tools in Python](#)
- Helped develop an embedded webpage used to control phased array electronics
- High sensitivity electronics testing (noise, high power, etc.)

Colorado State University

Fort Collins, CO

Graduate Research Assistant — Thesis

- Design, schematic, and layout of a modular, X-band, FMCW drone radar
- [Signal processing / data visualization](#) for Analog Device's FMCW Phaser Radar Board
- Presented the work at the IEEE International Phased Array Symposium
- Phaser board work was presented by ADI at IMS 2022 and IEEE-APS 2022

CHILL Radar Lab

Greeley, CO

Lab Assistant

- Built a real-time radar display to [easily display and animate radar data](#)
- Built a server to [host and display real-time and historical radar data using PostgreSQL, Flask, Python](#)
- Performed characterizations and measurements of radar equipment

Colorado State University — ECE Department

Fort Collins, CO

Learning Assistant

- [Helped students understand electrical engineering fundamentals](#)
- Wrote exam/homework questions to test understanding of core concepts

Engineering Technology Services

Fort Collins, CO

Lab Assistant

- Built operating system images for mass deployment across the network of engineering workstations
- Managed \$300,000 in lab equipment for engineering network of 240 workstations and devices
- Maintained a 98% average customer rating for lab development, advanced support, and operational quality

EDUCATION

Colorado State University

Fort Collins, CO

M.S. Electrical Engineering — GPA: 3.9

Thesis - Design, Deployment, and Cost Considerations for DARMA; A Low-Cost and Lightweight FMCW Radar

Colorado State University

Fort Collins, CO

B.S. Electrical Engineering — GPA: 3.47

PROJECTS

Animated RF and Radar Educational Videos

- **Made high-quality animations using Manim** to teach others about Radar and RF engineering
- Videos available at @MarshallBrunerRF on YouTube
- "What is FMCW Radar and why is it useful?" - Completed
- "But how do you actually build an FMCW Radar?" - Finalizing Animations

FMCW Radar Bootcamp

- Developed **signal processing software and a presentation for an FMCW radar** to present at the 2022 AESS Radar Bootcamp

Algorithmic Stock Trading Server

- Task scheduling system (backfilling data into PostgreSQL database, computing custom indicators, run entry / exit strategy)
- Containerized using docker
- Command-line interface for checking status, getting recent trades, etc.
- **Analysis / forecasting using Pandas, Numpy, PyTorch**
- Discord bot for general notifications / monitoring
- Dashboard using Plotly Dash and a Flask API

Open Source Contributions

- NuShell — A new type of shell
- Polars — Dataframes powered by a multithreaded, vectorized query engine, written in Rust

5G Electronics Senior Design Project

- Simulated, assembled, and measured a full 5G receiver system
- Created teaching documentation for a 5G transmitter and receiver for FR1 and FR2 frequency bands
- Experience using FieldFox, EMPro, ADS, SystemVue

Fuzzy Logic Clutter Filter

- Designed a radar clutter filter using machine learning

Environment-Mapping Car

- Built and programmed a battery-powered car with attached ultrasonic sensor for mobile mapping of the surrounding environment

SKILLS & ABILITIES

Languages: **Python**, Rust, C, Bash, SQL, Matlab, NuShell, JavaScript

Design / Simulation Tools: AWR Microwave Office, HFSS, KiCad, QucsStudio, LTSpice, Cadence

Technologies / Tools: Benchtop measurement equipment (VNA, Oscilloscope, etc.), Git, L^AT_EX, Linux, Docker, PostgreSQL, Flask

Communication: Excellent Technical Communication and Presentation Skills, Experience Presenting PDR/CDRs, Technical Presentation Formatting